IGCSE 0607 Practice Questions — E1.13 — Combined Questions & Markscheme

Using a calculator

Total Questions: 12 — **Total Marks:** 58 Marks

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- **E1.13** Using a calculator
  → **Paper 4 (Calculator)** *(12 Qs)*

1 Calculate the value of

$$\frac{\sqrt{15.2 - 3.4^2}}{4.1 \times 0.25}$$

Write down your full calculator display.

..... **[2]**

2 Work out the value of

$$\frac{\sqrt[3]{25.6 - 3.2^2}}{1.8 \times 0.45}$$

(a) Write down all the figures on your calculator display.

..... **[2]**

(b) Write your answer to part (a) correct to 3 decimal places.

..... **[1]**

3 Consider the following calculation.

(a) Work out the value of

$$\frac{\sqrt{4.1^2 + 5.2}}{2.3^3 - 1.4}$$

Write down all the figures on your calculator display.

..... **[2]**

(b) Give your answer to part (a) correct to 3 significant figures.

..... **[1]**

4 **(a)** Work out the value of

$$\frac{4.5^2 - \sqrt{18.2}}{3.1 + 0.8^3}$$

..... **[2]**

(b) Calculate the value of

$$2\sin(48^\circ) - \cos(35^\circ)$$

Give your answer correct to 3 decimal places.

..... **[2]**

5 **(a)** Use your calculator to find the value of

$$3.5^4 + \sqrt[3]{87.6}$$

Give your answer correct to 2 decimal places.

..... **[2]**

(b) Calculate the value of x when

$$x = \frac{2 \times \sin(43°)}{\cos(18°)}$$

Give your answer correct to 3 significant figures.

$x =$ ..... **[2]**

6 A student uses a calculator to solve three different problems.

(a) The answer to a problem about money is displayed on the calculator as 8.4.
Write this amount appropriately in dollars and cents.

..... **[1]**

(b) The answer to a problem about time is displayed on the calculator as 5.15 hours.
Write this time appropriately in hours and minutes.

hours ..... , minutes ..... **[1]**

(c) A car travels 280 km at an average speed of 85 km/h. Calculate the time taken for this journey.
Give your answer in hours and minutes, correct to the nearest minute.

hours ..... , minutes ..... **[3]**

7 **(a)** A circular swimming pool has a radius of 4.2 metres.
Calculate the area of the pool.
Give your answer correct to the nearest square metre.

..... m^2 **[2]**

(b) A train journey takes 4 hours and 23 minutes.
The average speed of the train is 78 km/h.

Calculate the distance travelled by the train.
Give your answer correct to 1 decimal place.

..... km **[4]**

8 A plumber charges a fixed fee of \$45 plus an hourly rate of \$32.50.

(a) The plumber works on a job for 3 hours and 40 minutes.
Calculate the total cost of this job.
Give your answer to the nearest cent.

\$ ..... **[3]**

(b) On another job, the total cost charged by the plumber is \$230.25.
Calculate the time the plumber worked, giving your answer in hours and minutes.

hours, minutes **[3]**

9 A water tank is in the shape of a sphere with radius 1.4 m.
The tank is completely full of water.

(a) Work out the volume of water in the tank.
Write down all the figures on your calculator display.

..... m^3 **[2]**

(b) Water leaks from the tank at a constant rate of 0.035 cubic metres per minute.
Calculate the time taken for the tank to become completely empty.
Do not round any intermediate values during your working.
Give your answer in hours and minutes, correct to the nearest minute.

hours ..... , minutes ..... **[4]**

10 **(a)** Calculate the exact value of

$$\left(3.2 \times 10^{5}\right) \times \left(4.5 \times 10^{-2}\right)$$

Give your answer in standard form.

..... **[2]**

(b) Work out the value of

$$\frac{1}{\sqrt{2\pi}} \times e^{-0.5^2}$$

Give your answer correct to 4 decimal places.

..... **[2]**

(c) A formula is given by $T = 2\pi\sqrt{\frac{l}{g}}$.

Find the value of l when $T = 1.5$ and $g = 9.81$.

Give your answer correct to 2 decimal places.

$l =$ ..... **[2]**

11 **(a)** Calculate

$$\frac{3.4 \times 10^5 + 2.7 \times 10^4}{8.1 \times 10^{-3}}$$

Give your answer in standard form.

..... **[3]**

(b) A solid metal cone has a base radius of r cm and a height of h cm.
$r = 5.2$ correct to 1 decimal place.
$h = 12.4$ correct to 1 decimal place.
Calculate the upper bound for the volume of the cone.
The volume, V, of a cone with radius r and height h is $V = \frac{1}{3}\pi r^2 h$.

..... cm^3 **[3]**

12 The volume, V, of a spherical cap is given by the formula

$$V = \frac{1}{3}\pi h^2(3R - h)$$

where R is the radius of the sphere and h is the height of the cap.

(a) Calculate V when $R = 8.45$ cm and $h = 3.2$ cm. Give your answer correct to 1 decimal place.

$V =$ ..... cm^3 **[2]**

(b) Rearrange the formula to make R the subject.

$R =$ ..... **[3]**

(c) A different spherical cap has volume $V = 500$ cm^3 and height $h = 5.6$ cm. Calculate the radius, R, of this sphere.
Give your answer correct to 3 significant figures.

$R =$ ..... cm **[2]**

1 Numerator $\approx 1.907878...$, Denominator $= 1.025$ **[M1]**

1.861344784 **[A1]**

[Back to Question] *[Total Marks: 2]*

2 **(a)** $3.068822...$ **[B2]**

If 0 scored, allow [B1] for seeing 15.36 or $2.485...$ or 0.81

(b) 3.069 [B1] (strict FT from their (a) if given to at least 4 d.p.)

[Back to Question] *[Total Marks: 3]*

3 **(a)** Numerator $\sqrt{22.01} = 4.69148\ldots$ or Denominator $12.167 - 1.4 = 10.767$ seen **[M1]**

0.435727839 **[A1]**

(b) 0.436 **[B1]**

[Back to Question] *[Total Marks: 3]*

4 **(a)** $4.423...$ or $\dfrac{15.98...}{3.612}$ **[M1]**

4.42 or 4.424 **[A1]**

(b) $1.4862... - 0.8191...$ or $0.6671...$ **[M1]**

0.667 **[A1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** $150.0625 + 4.4412...$ or $154.5037...$ **[M1]**

154.50 **[A1]**

(b) $\dfrac{1.3639...}{0.9510...}$ or $1.434...$ **[M1]**

1.43 **[A1]**

[Back to Question] *[Total Marks: 4]*

6 **(a)** 8.40 **[B1]**

(b) 5 hours 9 minutes **[B1]**

(c) $280 \div 85$ **[M1]**

$3.294...$ **[A1]**

3 hours 18 minutes **[A1]**

[Back to Question] *[Total Marks: 5]*

7 **(a)** $\pi \times 4.2^2$ or 55.417... **[M1]**

55 **[A1]**

(b) $23 \div 60$ or 0.3833... seen **[M1]**

Time $= 4 + \dfrac{23}{60}$ or 4.3833... hours **[M1]**

Distance $= 78 \times 4.3833...$ **[M1]**

341.9 **[A1]**

[Back to Question] *[Total Marks: 6]*

8 **(a)** Time $= 3 + \dfrac{40}{60}$ or $\dfrac{11}{3}$ hours **[M1]**

Cost $= 45 + 32.50 \times \dfrac{11}{3}$ **[M1]**

164.17 **[A1]**

(b) $230.25 - 45 = 185.25$ **[M1]**

$\dfrac{185.25}{32.50} = 5.7$ hours **[M1]**

5 hours 42 minutes **[A1]**

[Back to Question] *[Total Marks: 6]*

9 **(a)** $\dfrac{4}{3}\pi(1.4)^3$ **[M1]**

11.49404... **[A1]**

(b) their 11.494... $\div$ 0.035 **[M1]**

328.4... or 328 (minutes) **[A1]**

their 328.4... $\div$ 60 (converting to hours) **[M1]**

5 hours 28 minutes **[A1]**

[Back to Question] *[Total Marks: 6]*

10 **(a)** 14.4×10^3 or 14400 **[M1]**

1.44×10^4 **[A1]**

(b) $\dfrac{1}{2.5066...} \times 0.7788...$ or $0.398942... \times 0.778800...$ or 0.310696... **[M1]**

0.3107 **[A1]**

(c) $1.5 = 2\pi\sqrt{\dfrac{l}{9.81}} \implies \dfrac{1.5}{2\pi} = \sqrt{\dfrac{l}{9.81}} \implies 0.2387... = \sqrt{\dfrac{l}{9.81}}$ **[M1]**

$l = 9.81 \times (0.2387...)^2 \implies l = 0.55909...$ **[M1]**

0.56 **[A1]**

[Back to Question] *[Total Marks: 6]*

11 **(a)** Numerator $= 340000 + 27000 = 367000$ or 3.67×10^5 **[M1]**

$3.67 \times 10^5 \div (8.1 \times 10^{-3}) = 45308641.98...$ **[M1]**

4.53×10^7 or 4.531×10^7 **[A1]**

(b) Upper bound for $r = 5.25$ and Upper bound for $h = 12.45$ **[M1]**

$V = \frac{1}{3} \times \pi \times 5.25^2 \times 12.45$ **[M1]**

359.34... or 359 **[A1]**

[Back to Question] *[Total Marks: 6]*

12 **(a)** $\frac{1}{3} \times \pi \times 3.2^2 \times (3 \times 8.45 - 3.2)$ **[M1]**

237.5 **[A1]**

(b) $\frac{3V}{\pi h^2} = 3R - h$ **[M1]**

$\frac{3V}{\pi h^2} + h = 3R$ **[M1]**

$R = \frac{V}{\pi h^2} + \frac{h}{3}$ or $R = \frac{3V + \pi h^3}{3\pi h^2}$ **[A1]**

(c) $\frac{500}{\pi \times 5.6^2} + \frac{5.6}{3}$ **[M1]**

6.94 **[A1]**

[Back to Question] *[Total Marks: 7]*